Ch. 5 Monte Carlo Methods

I. Intro
A. Unlike DP, do not assume complete knowledge of env.
1. Learn from experience only
2. Do not need probability distributions
3. Can still get optimal behavior based on sample s, a, & r
B. MC is nonstationary
1. Return from taking an action depends on later actions (unlike bandit)
2. Can use GPI, like DP, to solve nonstat.
3. Unlike DP w/ computed value, here we learn value with samples returns

II. MC Prediction
A. If value is the expected return over next states, MC suggests that averaging over many returns after an action will converge to the value of a state, no probabilities needed.
B. Given finding $V_\pi(s)$ and given a set of episodes from π interacting with s...
1. Each occurence of s in episode is a visit
a. a first visit is the first occurence of s in e
2. Two MC methods, first-visit & every-visit.
a. first-visit estimates $V_\pi(s)$ averaging returns following first
b. every-visit ~~~~~ averages following all visits
c. both converge to $V_\pi(s)$
C. Consider Blackjack, knowing exact probabilities to make a value decision based on what you and the dealer has is impractical
D. The independent nature of MC allows for value estim. of single states; Episodes can be made and evaluated from states of interest

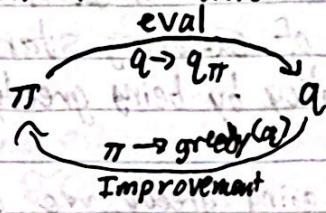
III. MC Action Val. Estim.
A. With no model, finding V_* requires estimating value for each action and ∴ is not sufficient, can't find best value of reward & next state. We must find q_* in MC
B. Instead of state visits, use state-action pair visits
1. Guaranteed to happen only in start in state-action pair exploring starts and stochastic non-zero probability policies to select all pairs.
2. Important for maintaining exploration

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IV. MC Control

A. Approxim. optimal policy is much the same in MC as DP by following generalized policy iteration (GPI)

1. Keep an approx. policy & value function s/t value is updated to closely approx. the policy & policy is updated based on the value

2. 

OR, in terms of starting w/ arb. π_0 and ending optimal:

$$\pi_0 \xrightarrow{E} q_{\pi_0} \xrightarrow{I} \pi_1 \xrightarrow{E} q_{\pi_1} \xrightarrow{I} \pi_2 \xrightarrow{E} \cdots \xrightarrow{I} \pi_* \xrightarrow{E} q_*$$

3. Assuming ∞ episodes & exploring starts, MC computes each q_{π_k} exactly for arbitr. π_k

B. Policy improvement is made w/ greedy policy regarding value and, since we have <u>action-value</u> pairs, the greedy policy is that which, for every state $s \in S$, determinis. chooses action with MAX action-value:

1. $\pi(s) = \arg\max_a q(s,a)$
2. Thus the improvement rule looks like:

$$\begin{aligned} q_{\pi_k}(s, \pi_{k+1}(s)) &= q_{\pi_k}(s, \arg\max_a q_{\pi_k}(s,a)) \quad \text{(1.)} \\ &= \max_a (q_{\pi_k}(s,a)) \quad \text{(2.)} \\ &\geq q_{\pi_k}(s, \pi_k(s)) \\ &= V_{\pi_k}(s) \end{aligned}$$

1. Transition is saying same deal, get max action to maxim. val
2. The action becomes policy given it is an improvement

3. Formula ensures uniform improvement or remaining at best π.
4. W/o ∞ episodes & exploring starts will be done later

C. Monte Carlo ES (exploring starts) can be used to solve for optimal policy w/o a model by averaging returns of s, a pairs and for each state in the episode setting the policy to the action of maximum return. (Figure 5.4)

V. MC Control w/o Exploring Starts

A. Having ES is unlikely, and resulting solutions fall into <u>on-policy</u> and <u>off-policy</u> methods.

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1. On-policy focus on improving & eval. the policy making decisions.
2. Off-policy eval/improve a policy different from current data-generating π

B. On-policy methods are generally soft, meaning $\pi(a|s) > 0 \ \forall \ s \in S$ & $a \in A(s)$
1. ε-greedy falls into category, always having a small prob. ε to select an action at random $\left(\frac{\varepsilon}{|A(s)|}\right)$
2. ε-greedy considered ε-soft, where $\pi(a|s) \geq \frac{\varepsilon}{|A(s)|}$ $\forall \ s \in S$ & $a \in A(s)$ for some $\varepsilon > 0$
3. Using this fact, we can get rid of exploring starts
   a. With ES, we can't improve policy by being greedy since not all options are explored
   b. However, w/ GPI, instead of going greedy, we can just move towards greedy and ∴ have constant improvement of π
      Proof in 5.2 Equation
4. Ultimately a MC control can be written not relying on environment and incorporating ε to visit all possibilities. (Figure 5.6)

VI. Off-Policy Pred. via Importance Sampling
A. Suppose all the episodes we have are not of our policy π but a different policy μ and we want to learn π.
1. π is the target policy, μ is the behavior policy
2. To get π, μ is required to at least occasionally visit all π's actions (aka $\pi(a|s) > 0$ implies $\mu(a|s) > 0$). This is coverage
3. Typically μ is stochastic/exploratory (ε-greedy) and π is deterministic

B. Importance Sampling estimates expected vals under one distr. given another
1. The importance-sampling ratio weights returns based on how likely the relative trajectories occur under the target and behavior policy
2. ∴, given start state S_t, prob of state-action trajectory is:

$$\prod_{k=t}^{T-1} \pi(A_k|S_k)\, p(S_{k+1}|S_k, A_k) \quad (1)$$

multiply over action-state trajectory

$\pi(A_k|S_k)$: prob action A_k will be chosen from S_k under π

$p(S_{k+1}|S_k, A_k)$: prob A_k being chosen in S_k results in expected state S_{k+1}

T-1 is terminal
k=t means start from S_t, state to eval

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3. Thus, under ①, the relative trajectory probability for target & behavior policies is

$$\rho_t^T = \frac{\prod_{k=t}^{T-1} \pi(A_k|S_k)\, p(S_{k+1}|S_k, A_k)}{\prod_{k=t}^{T-1} \mu(A_k|S_k)\, p(S_{k+1}|S_k, A_k)} = \boxed{\prod_{k=t}^{T-1} \frac{\pi(A_k|S_k)}{\mu(A_k|S_k)}}$$

NOTE: Even though we rely on unknown $p(s'|s,a)$ values which would require environment, they cancel, so the unknown values for transition probabilities don't matter

4. ∴, to find estimated value of $s \in S$, consider set of all time steps (where s is visited) as $\mathcal{T}(s)$, $T(t)$ denotes first time of termination following t, and $G(t)$ denotes return following t through $T(t)$. THEN:

$$V(s) = \frac{\sum_{t \in \mathcal{T}(s)} \rho_t^{T(t)} G_t}{|\mathcal{T}(s)|}$$

($V(s)$: value of s; averaging over all s occurrences; ρ modifies the observed value G_t from the behavior by how important it is to choose a given s comparatively to π.)

This is called ordinary importance sampling ← (OIS)

5. Weighted importance sampling w/ weighted avg.:

$$V(s) = \frac{\sum_{t \in \mathcal{T}(s)} \rho_t^{T(t)} G_t}{\sum_{t \in \mathcal{T}(s)} \rho_t^{T(t)}}$$

↖ (WIS) vs. (OIS)

6. Both tend to be unrepresentative of $v_\pi(s)$ with few occurences,
   i. Weighted, at 1 visit, is biased to μ b/c $\rho_t^{T(t)}$ cancels, leaving G_t
   ii. Ordinary, at 1 visit, represents π but can be extreme based on $\rho_t^{T(t)}$
   iii. Usually weighted is preferred b/c of less variance

VII. Incremental Implementation

A. MC pred. methods don't need to have all episodes upfront, they can update per-episode
   1. Ch. 2 methods of averaging rewards apply now to returns
   2. In off-policy, OIS returns are scaled by $\rho_t^{T(t)}$ and averaged, but WIS is different

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B. In WIS, if we have returns $G_1, G_2, \ldots, G_{n-1}$ w/ corresponding weights in $W_i = \rho_t^{T(t)}$ for state S:

1. The estimate becomes: $V_n = \frac{\sum_{k=1}^{n-1} W_k G_k}{\sum_{k=1}^{n-1} W_k}$ ← cumsum of weights

and kept up to date when obtaining G_n

2. Thus, to update this sum to new V_{n+1} and C_{n+1}...

$$V_{n+1} = V_n + \frac{W_n}{C_n}[G_n - V_n], \quad n \geq 1 \quad \text{WHERE } C_{n+1} = C_n + W_{n+1}$$

(V_n: current value; $\frac{W_n}{C_n}$: weight all value that came before this step)

and $C_0 = 0$

V_1 = arbitrary

VIII. Off-Policy MC

A. Already defined earlier, but advantage is that the behavior policy can be allowed to sample all actions while the target policy is greedy + deterministic.

B. Behavior Policy must be soft ($p > 0$ that all actions can be chosen)

C. Example alg Fig 5.10 helpful to see how incremental behavior updates are used to update the greedy policy with best actions

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